mitsue_village_re_plan_alignment

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Alignment with the Mitsue Village Renewable Energy Plan

How the Mitsue Project delivers the village's own official decarbonization strategy

Prepared for: internal strategy, grant applications, and village government engagement
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1. What this document is

In January 2025 (Reiwa 7), Mitsue Village published its official **"Plan for Maximum Introduction of Renewable Energy"** (御杖村再エネ導入最大化計画概要版), produced by the village's Policy Promotion Division and **funded by a Ministry of the Environment grant** (FY2023 supplementary budget, "regional decarbonization / maximum renewable-energy introduction planning support project," administered via the Association for Regional Circular and Ecological System / 地域循環共生圏).

This is not a think-tank paper or an outside proposal. It is **the village's own statutory decarbonization strategy**, aligned with the village's Long-Term Comprehensive Plan and its Global Warming Countermeasures Implementation Plan, running from Reiwa 6 (2024) to Reiwa 32 (2050).

The strategic significance is simple and large:

> **The Mitsue Project is not asking the village to adopt a new idea. It is offering to be the implementation and operating vehicle for the plan the village has already written, funded, and committed to.**

That reframing changes how we pitch to the mayor, how we apply for funding, and how we describe ourselves to partners. The rest of this document sets out the alignment, the funding unlock, the citable figures, and the per-audience messaging.

> **Translation note.** The village's English machine translation (in the repository) is heavily garbled — the village name appears as "Mitsu / Otsuke / Gose / Mikawa / Gozumura," all of which are 御杖村 (Mitsue-mura, Uda-gun, Nara). A clean

> English translation of the key pages is at `Docs extra/mitsue_files from Village hall/mitsue_village_re_plan_clean_translation_en.md`.

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2. The core thesis: we are the vehicle, not a competing plan

The village plan sets ambitious goals but, as a municipal policy document, it largely lacks concrete delivery mechanisms, capital, and an operator. The Mitsue Project supplies exactly those missing pieces — and our energy logic (biomass CHP **primary**, solar + battery + EV charging complementary) maps onto the plan's stated priorities with very little friction.

Critically, the plan already endorses, in its own words, almost every technology in our design: small-scale hydropower, rooftop and carport solar, **perovskite** solar, storage batteries, **EV charging at public facilities as a priority**, **V2H**, **VPP using IoT/blockchain**, off-grid/distributed energy for disaster resilience, wood biomass and wood stoves, and **J-Credits from forest carbon**. We are not introducing unfamiliar technology; we are operationalizing a list the village has already published.

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3. Mapping — village plan → Mitsue Project

Village RE plan says it wants…	The Mitsue Project delivers…	Plan reference
An "autonomous & distributed energy society for **disaster preparedness**"; supply power to public buildings/clinics when the grid fails (off-grid capability)	Biomass CHP (24/7 baseload) + solar + battery → blackout resilience for critical village facilities	Basic Policy 2.2
An explicit indicator target of **"one resilient site within the village"** using renewables + storage + EV — currently **zero**	The data center site **is** that resilient distributed-energy site	Indicators table (p.18)
EV chargers at "town hall and roadside stations **as a priority**"; EV/PHEV/FCV promotion (transport = 46% of emissions, the #1 source)	EV charging network anchored to local generation, scaling with adoption	Basic Policy 3.1
EVs/PHEVs as "mobile storage batteries"; **V2H** to use stored EV power in buildings	EV + V2H integration evaluated in the energy feasibility study	Basic Policy 2.2
Local production for local consumption of	A locally owned generation + consumption	Key Initiative 1

Village RE plan says it wants…	The Mitsue Project delivers…	Plan reference
renewable energy unique to the village's geography	system (CHP + solar) feeding a local offtaker	
Small-scale hydropower from water-supply intake points	In-scope option for the Phase 1 energy feasibility study	Key Initiative 1
Forest CO_2 absorption, planned thinning, clarified forest ownership/boundaries, and a **J-Credit** system for forest carbon	Sugi thinning (which fuels our biomass CHP) + native reforestation = the concrete mechanism that produces the absorption and the J-Credits the plan wants	Basic Policy 5.3; Key Initiative 2
VPP (Virtual Power Plant) using IoT/blockchain to stabilize the grid and level demand	A controllable compute load + battery is a natural VPP node	Basic Policy 5.2
Branding "**eco rural living**" to attract young residents and secure forestry successors; raise civic pride	Local jobs (forestry, facility ops, EV/energy), an open replicable model, and international visibility	Key Initiative 3
ZEB/ZEH, high-efficiency equipment, energy-saving renovation	The repurposed-facility data center and Koryukan renovation can be specified to these standards	Basic Policy 1

The two highest-value rows are the resilient-site target and the forest/J-Credit mechanism:

- **The "one resilient site" target.** The plan sets a target of exactly one resilient distributed-energy site in the village and currently has none. Our data center, with on-site generation and battery backup, can credibly *be* that site — letting the village report progress against its own published indicator.
- **The forest / J-Credit mechanism is uniquely ours.** The plan repeatedly states it *wants* forest CO_2 absorption, planned thinning, and J-Credit revenue — but it has no operator, no capital, and no offtaker for the thinnings. **Our biomass CHP is that mechanism**: it consumes sugi thinnings, which funds and accelerates the thinning, which clears the way for native broadleaf restoration, which produces the certified forest carbon. Energy output and ecological restoration are the same lever. This is why biomass CHP remains the project's **primary** energy source and must never be demoted to fit the plan's solar-led framing.

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4. The funding unlock — this is the most important section

The plan's last page reveals the funding thread that matters most. The plan itself was produced under the Ministry of the Environment's **planning-support grant** (「地域脱炭素実現に向けた再エネの最大限導入のための計画づくり支援事業」). That is **step one of a three-step national funding ladder** — and Mitsue has already completed it.

The MoE decarbonization funding ladder

Step	Programme	Status for Mitsue	What it funds
1. Plan	計画づくり支援事業 (planning support)	✅ **Done** — this is what produced the RE plan	The strategy document itself
2. Build	**地域脱炭素移行・再エネ推進交付金** (Regional Decarbonization Transition & RE Promotion Grant)	Eligible — needs a multi-year 交付金事業計画 submitted **by the village**	Solar, batteries, **private wire (自営線)**, EV, energy-saving equipment — i.e. our hardware
3. Premier	**脱炭素先行地域** (Decarbonization Leading Area) selection	Aspirational — the plan's 2045 CN target positions for it	RE + base infrastructure + efficiency at the highest support tier

Why this is a major unlock for our budget

- **Subsidy rate.** The 交付金 covers **2/3** of eligible cost as standard, rising to **3/4** for storage batteries and private-wire (自営線) when the municipality has a low financial-capacity index **and** the installation site is in an underpopulated (過疎) area. **Mitsue qualifies on both counts.** A large share of our solar + battery + private-wire + EV capex could therefore be subsidized at 2/3–3/4.
- **It routes through the village, by design.** The grant is paid to the **municipality** (地方公共団体), and the planning-support programme explicitly funds "building the implementation and operating structure for regional renewable-energy projects carried out through **public-private partnership (官民連携)**." That is precisely the slot the Mitsue Project fills: **the village brings the plan and the grant channel; we bring the operator, the capital stack, and the anchor offtaker.** This is the concrete, financial reason village partnership is our funding strategy — not merely goodwill.
- **It strengthens, not replaces, our existing stack.** This becomes the backbone of **Layer 2 (Government grants)** in our funding model. It does not change the biomass CHP economics (CHP is funded through forestry/biomass and METI/NEDO routes), but it can substantially de-risk the complementary solar + battery + EV + private-wire investment that the data center and blackout-resilience case depend on.

Implication for the EVM / funding gap

Our current baseline shows a **¥28M–¥53M funding gap** against BAC ¥220M. A successful village-led 交付金 application at 2/3–3/4 on the solar/battery/EV/private-wire portion is a credible, named path to closing a meaningful part of that gap during Phases 2–3 — and it is more concrete than "additional government grants" stated generically. This should be reflected explicitly in the EVM plan, the funding flowchart, and the implementation plan's Layer 2.

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5. Citable official figures (from the village's own plan)

Use these — they are the village's published numbers, which carries more weight than third-party estimates.

Figure	Value	Use it to…
Village CO_2 emissions (2021)	**9 thousand t-CO_2**	size the problem with the village's own number
Baseline emissions (2013 / Heisei 25)	13.2 thousand t-CO_2	show the reduction trajectory
Transport share of emissions	**46% — the largest sector**	justify EV charging as the highest-impact intervention
Residential / Industrial / Other-business / Waste	20% / 18% / 13% / 3%	sector context
Forest cover	**~90% of village area**	ground the forestry + J-Credit case
Warming observed near the village	**+2.1°C over 42 years**	the local climate stakes
CO_2 reduction target	**60% by 2030** vs 2013	the near-term goal we help hit
Carbon-neutrality target	**2045** (Reiwa 27) — 5 years ahead of national	the village's ambition, which we amplify
EV/hybrid adoption	30% (2024) → 58% (2030) → **100% (2045)**	our EV infrastructure serves this curve
Home solar	4.1% → 6.0% → 14.0%	complementary solar context
Forest CO_2 absorption	0.3 → 1.8 → **2.7 kt/yr**	our reforestation contributes here
Resilient renewable+storage+EV sites	now: **0** → target: **1 site**	our data center can be the "1"

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6. How to use this, by audience

- **Village mayor / government.** Lead with: "This project is how the village executes its own renewable-energy plan." Emphasize the resilient-site target it lets them report, the EV-charging priority it fulfills, and the 交付金 funding that flows through the village. Pair with the existing mayor talking points and the "5 enterprises in 5 years" alignment.
- **Government grant applications.** Cite the plan as the policy basis; position the project as the 官民連携 operating partner for the village's 交付金 事業計画; use the official emissions/target figures; map deliverables to the plan's Basic Policies and indicators.
- **Foundations & corporate partners.** Use the alignment as third-party validation that the project is embedded in official policy, not speculative — de-risking the philanthropy/CSR case.

- **International outreach (HIGHRESO, Ito network, Dutch consulate).** Use the clean English translation + this alignment as evidence of official local commitment and a clear public-policy tailwind.

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7. What this does not change

- **Biomass CHP stays primary.** The village plan leads with solar/EV/small-hydro and mentions wood biomass only lightly (wood stoves). Our project leads with biomass CHP because it is the engine of reforestation. We align *to* the plan without inverting our own energy hierarchy. Our framing: we deliver the plan's solar/battery/EV/resilience targets **and** we supply the forest-carbon mechanism the plan wants but cannot deliver alone.
- **The 25-year horizon stays.** The plan runs to 2050; our horizon and phase gates are unchanged.
- **No new commitments to the village.** Alignment is a positioning and funding strategy, not a promise to fund the village's plan ourselves.

Sources

- 御杖村再エネ導入最大化計画（概要版）, Mitsue Village, Jan 2025 — `Docs extra/mitsue_files from Village hall/20250130saienekeikakugaiyou_translated_eng-1.pdf`
- MoE — 地域脱炭素推進交付金: https://policies.env.go.jp/policy/roadmap/grants/
- MoE — 脱炭素先行地域: https://policies.env.go.jp/policy/roadmap/preceding-region/
- MoE — 再エネ最大限導入のための計画づくり支援事業: https://www.env.go.jp/policy/post_169.html
- MoE — 地域脱炭素移行・再エネ推進交付金 実施要領（補助率 2/3・3/4 条件）: https://www.env.go.jp/content/900470616.pdf

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